EXPERIMENT 1: CAESAR CIPHER

1. Aim

To write and execute a C program to implement the Caesar cipher by replacing each letter of the plaintext with another letter shifted by a key value between 1 and 25.

2. Algorithm

1. Start the program.
2. Read the plaintext and the key value.
3. Check whether the key is between 1 and 25.
4. For each alphabetic character, shift its position by the key value.
5. Apply modulo 26 to wrap around the alphabet.
6. Display the encrypted text.
7. Stop the program.

CODE:

```
import subprocess
code = r'''
#include <stdio.h>
int main()
{
    char text[] = "HELLO";
    int k = 3;
    int i;
    printf("Plaintext: %s\n", text);
    printf("Key: %d\n", k);

    for (i = 0; text[i] != '\0'; i++)
    {
        if (text[i] >= 'A' && text[i] <= 'Z')
        {
            text[i] = (text[i] - 'A' + k) % 26 + 'A';
        }
    }
    printf("Ciphertext: %s\n", text);

    return 0;
}
'''
# Save the C program
with open("caesar.c", "w") as f:
    f.write(code)

# Compile the C program
```

```
# Save the C program
with open("caesar.c", "w") as f:
    f.write(code)

# Compile the C program
compile_result = subprocess.run(
    ["gcc", "caesar.c", "-o", "caesar"],
    capture_output=True,
    text=True
)

if compile_result.returncode != 0:
    print("Compilation error:")
    print(compile_result.stderr)
else:
    # Execute the program
    result = subprocess.run(
        ["./caesar"],
        capture_output=True,
        text=True
    )
    print(result.stdout)

    # Save output for the lab record
    with open("caesar_output.txt", "w") as f:
        f.write(result.stdout)
    print("C program saved as caesar.c")
    print("Output saved as caesar_output.txt")
```

OUTPUT:

```
Plaintext: HELLO
Key: 3
Ciphertext: KHOOR
```

RESULT:

Thus, the C program for the Caesar cipher was implemented and executed successfully, encrypting the plaintext HELLO into the ciphertext KHOOR using a key value of 3.